

Deepak Mallampati

PLATFORM / MLOPS ENGINEER

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US-based, F-1 OPT, open to sponsorship

PROFESSIONAL SUMMARY

Platform / MLOps engineer with 4+ years building and operating production AI infrastructure across multi-cloud environments (AWS, Azure, GCP). Designed job orchestration and fault-tolerant batch pipelines that cut failed production runs by **60%**, stood up CI/CD and LLM evaluation/observability frameworks that catch regressions before release, and shipped APIs/microservices serving **100,000+ requests daily** with p95 latency reduced from 1.5s to 800ms. Strong in Kubernetes, Docker, Terraform, and reliability engineering - comfortable owning the full lifecycle from cluster operations to backend services.

CORE COMPETENCIES

Kubernetes & Docker • Terraform / IaC • CI/CD (Jenkins, Azure DevOps) • Job Orchestration • Observability & Monitoring • Multi-Cloud (AWS/GCP/Azure) • Python • Production Linux • MLflow / W&B • Reliability Engineering

MLOps & Cloud: AWS (EC2, S3, Lambda, Bedrock, SageMaker), Azure, GCP Vertex AI | Docker, Kubernetes | Terraform | CI/CD (Jenkins, Azure DevOps) | MLflow, Weights & Biases | Model Monitoring & Observability

Languages & Backend: Python, JavaScript/TypeScript, Java, SQL | FastAPI, React | Git

Data & Infra: PostgreSQL, MongoDB, Redis | Spark, Databricks | Kafka | Pandas, NumPy | Data Lineage & Governance

AI/ML Platforms: LLM eval (RAGAS, BERTScore) | RAG, vector search (FAISS, Pinecone, Weaviate) | Fine-tuning (LoRA/QLoRA, H100) | Drift detection

PROFESSIONAL EXPERIENCE

Progress Solutions

Jul 2025 - Present | USA

AI Engineer

- Built fault-tolerant **automation infrastructure** with scheduled batch orchestration and exponential-backoff retry logic, reducing failed production runs by **60%**.
- Established a structured **LLM evaluation and monitoring framework** (RAGAS, BERTScore, custom metrics) with latency and accuracy dashboards to surface regressions before release.
- Architected production-grade **agentic orchestration** on a conductor-subagent framework, decomposing objectives into context-scoped subagents - cutting token consumption by **42%** while sustaining task accuracy.
- Optimized inference cost and latency through prompt compression, semantic caching, and model routing, sustaining SLA targets while reducing per-query overhead.
- Engineered high-performance RAG pipelines over large-scale corpora with dense retrieval + cross-encoder re-ranking.

Vanguard

Jan 2024 - Jan 2025 | USA

AI Engineer Intern

- Built **12 APIs and microservices** powering new AI capabilities, reducing p95 latency from 1.5s to **800ms** and accelerating feature integration by 40%.
- Partnered with product, platform, and data engineering teams to deliver scalable LLM applications supporting **100,000+ requests daily** across 20+ internal teams.
- Developed and enhanced AI agents and backend services for Vanguard's internal AI platform, integrating with production data pipelines and observability tooling across 25+ AI agents.
- Optimized and evaluated GPT-4o, Claude Sonnet, and Gemini models, improving task success rates by 27% and informing platform model-selection decisions.

Kent State University

Oct 2023 - Jan 2024 | USA

ML & Gen AI Research Assistant

- Fine-tuned **Qwen3-27B** via **4-bit QLoRA** on an **NVIDIA H100**, managing GPU memory, quantization, and curated instruction datasets for controllable long-form generation.
- Built a privacy-preserving ML pipeline (k-anonymity, l-diversity, differential privacy) retaining 99% of baseline predictive performance.

Emerson

Jun 2022 - Apr 2023 | India

ML Trainee

- Built regression/classification models on operational pipeline-storage data for equipment-failure prediction and anomaly detection; engineered end-to-end preprocessing pipelines.

PROJECTS

career-ops

Autonomous AI job-search pipeline that scans listings, scores fit, and generates tailored applications overnight - orchestration, scheduling, and reliability built in.

MangaLens

Shipped Chrome extension (live on Chrome Web Store) delivering real-time AI translation across 29+ sites.

EDUCATION

M.S. in Computer Science - Kent State University, OH

2025 | GPA 3.70/4.0